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Dietary inflammatory index (DII) and attention deficit hyperactivity disorder: a cross-sectional study in US children and adolescents and Mendelian randomization study

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Abstract

Background Diet may impact the risk of attention deficit hyperactivity disorder (ADHD) via altering inflammation. However, no investigations have been conducted to evaluate the relationship between ADHD and the diet's inflammatory potential. The study sought to investigate the relationship between ADHD and the dietary inflammatory index (DII).

Methods We used data from the cross-sectional National Health and Nutrition Examination Survey 2001–2004 to investigate the relationship between DII and ADHD in 6481 children and adolescents aged 6–19 years. The relationship between ADHD and DII was investigated using weighted logistic regression and restricted cubic spline (RCS). We also explored the potential causal associations between ADHD and 187 types of food preferences using Mendelian randomization (MR).

Results Despite thorough multivariate correction, a significant association between lower DII scores and an elevated probability of ADHD persisted (OR = 0.91, 95% CI 0.82–0.99). Nevertheless, when stratifying the DII into quartiles (T1–T4) for analysis, no statistically significant correlation was seen between DII and ADHD ($P > 0.05$). RCS analysis indicated a U-shaped correlation between DII and ADHD, with an inflection point at approximately 2.9. Mediation analysis indicated that the Frailty Index partly mediated the association between DII and ADHD, representing about 5.49% of the overall impact. The findings from the univariate MR analysis revealed that fruit liking is significantly linked to a reduced risk of ADHD. Conversely, a liking for red meat, such as lamb, as well as a fondness for flavorings like onions, is significantly associated with an increased risk of ADHD. Reverse two-sample MR analyses revealed that genetic predisposition to ADHD significantly altered 37 food preferences (FDR < 0.05).

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Conclusions We observed a U-shaped association between the DII and ADHD risk. The Frailty Index may partially mediate (5.49%) the association between DII and ADHD. However, the cross-sectional design precludes causal inference. Fruit liking lowered ADHD risk, while red meat/onion liking increased risk. ADHD was associated with a generalized reduction in liking for a wide range of foods. Future studies should investigate the role of a moderate-inflammatory diet in ADHD prevention.

Key points

1. U-Shaped Association: A non-linear, U-shaped association was observed between the Dietary Inflammatory Index (DII) and ADHD risk in children and adolescents, indicating that both highly pro-inflammatory and anti-inflammatory diets may be associated with increased risk.

2. Potential Mediation: The Frailty Index partially mediated (5.49%) the association between DII and ADHD, suggesting that the diet-ADHD link may be partly explained by diet's impact on overall health status.

3. Specific Food Preferences: Mendelian randomization analysis suggested that genetic predisposition for liking fruit is associated with lower ADHD risk, while liking for red meat and onions is associated with higher risk. ADHD genetics also broadly influenced food preferences.

4. Clinical Implication: Promoting a balanced, anti-inflammatory diet (e.g., increasing fruit intake) could be a valuable component of preventive strategies and clinical management for ADHD in the pediatric population.

Keywords Dietary inflammatory index, Attention deficit hyperactivity disorder, Frailty Index, National health and nutrition examination survey, Food preferences, Mendelian randomization

Introduction

Over the last two decades, there has been a substantial rise in pediatric neuropsychiatric disorders [1], resulting in severe implications for the afflicted children, their families, and the public health system. Attention deficit hyperactivity disorder (ADHD) is the most prevalent neuropsychiatric illness that begins in childhood, with an estimated global incidence of about 5% among school-aged children [2], and boys exhibit more symptoms than girls [3]. Prior studies have shown a substantial correlation between ADHD and inflammation, with inflammation perhaps contributing to the genesis of ADHD [4]. A meta-analysis of 10 studies revealed that persons with ADHD had increased levels of IL-6 and markedly decreased levels of TNF-α [5]. Although prior studies have not reached a consensus on the changes in inflammatory markers among individuals with ADHD, they generally support the notion that immune system dysregulation is present in this population [6, 7]. Inflammation is a fundamental biological process that facilitates the connection between an organism and its environment, including dietary factors. Numerous dietary elements linked to child neurodevelopment are recognized for their pro-inflammatory properties (e.g., fatty products) and anti-inflammatory properties (e.g., fermented goods) [8]. The Dietary Inflammatory Index (DII) is an epidemiological tool designed to estimate the overall inflammatory potential of an individual's diet based on the association between dietary components and levels of specific inflammatory biomarkers reported in the literature [9–11]. It provides a composite measure that facilitates the assessment of diet as a whole in relation to inflammation [12]. Furthermore, emerging research into mechanisms such as the microbiota-gut-brain axis suggests that diet may influence neurodevelopment and behavior, potentially through modulation of systemic inflammation and gut microbiota [13]. Observational studies have linked various dietary patterns (e.g., high intake of fruits and vegetables vs. high intake of processed snacks) to differences in ADHD symptoms or related behavioral profiles [14–16]. However, the overall inflammatory potential of the diet, as an integrative measure, has not been thoroughly investigated in relation to ADHD diagnosis.

Although there are possible links between food, inflammation, and ADHD, the precise impact of dietary inflammation (assessed by DII) on ADHD is still ambiguous. This research seeks to investigate the relationship between diet-induced inflammation, measured by the DII, and ADHD using data from the National Health and Nutrition Examination Survey (NHANES). Additionally, we conducted a multi-omics Mendelian randomization (MR) study to investigate the effects of various food preferences on ADHD. Through the examination of this connection, we aim to provide significant insights into the diet-ADHD relationship, which could inform future research on potential dietary strategies for ADHD management.

Materials and methods

Study participants

The National Center for Health Statistics (NCHS) of the U.S. Centers for Disease Control and Prevention (CDC) conducts the NHANES, a population-based, cross-sectional survey that evaluates the nutritional status and overall health of the civilian, noninstitutionalized population in the United States. The data used in this analysis were taken from this survey. A sophisticated, stratified multistage probability sample design is used to choose the sample, which oversamples certain subpopulations to reflect the U.S. population across all age groups. A home interview was used to gather data about the study participants. In a mobile examination facility, a standardized physical examination was performed, which included a few medical and laboratory procedures. The NCHS institutional review board authorized the study's operations; minor children (those under the age of 17) gave written agreement to participate, and children who gave their consent in front of the interviewer and their parent or guardian [17]. Publicly accessible data for children and adolescents aged 6–19 who took part in two consecutive NHANES cycles (2001–2002 and 2003–2004) were utilized in this research. A random subsample of subjects aged ≥ 6 was subjected to DII exposure measurements. Based on the availability of exposure, outcome, and important covariate data, we chose study cycles and age range. First, we included 21,161 subjects from two cycles. Second, we excluded 13,962 participants who were younger than 6 years old or older than 19 years old. Then, we further excluded 699 participants with missing dietary data. Finally, we excluded 19 participants whose ADHD data were categorized as "unknown," "refused to answer," or "missing." After screening, the final study comprised 6481 eligible subjects in total (Fig. 1). The procedures were approved by the NCHS's ethical review board, and participants completed informed consent documents.

Study variables

Measurement of diagnosed ADHD

Consistent with the primary outcome variable in prior NHANES studies [18, 19], we assessed ADHD diagnoses among children and adolescents using information derived from the National Institute of Mental Health (NIMH) Computerized Diagnostic Interview Schedule for Children, Version IV (C-DISC IV) framework. The questions in the diagnostic interview were structured according to the diagnostic criteria of the Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV) and the International Classification of Diseases, Tenth Revision (ICD-10). In the NHANES cycles from 2001 to 2004, ADHD diagnosis was obtained solely through the parent informant/self-report interview (DISC-P). It is important to clarify that although the questions were aligned with the C-DISC IV framework, the actual assessment in NHANES was based on parent-reported clinician diagnosis rather than a direct structured clinical interview. Specifically, parents were asked: "Has a doctor or health professional ever told [child] that he/she has attention deficit disorder?" [20].

DII exposure assessment

We used the Dietary Inflammatory Index (DII) to assess the overall inflammatory potential of participants' diets. The DII was developed by Shivappa et al. as a scoring system based on the effects of dietary components on six inflammatory biomarkers (IL-1β, IL-4, IL-6, IL-10, TNF-α, and CRP), originally comprising 45 dietary parameters (Table S1-1). As stated by Shivappa et al., the DII score remains valid and retains its predictive value for diet-related inflammation even when fewer than 30 parameters are used [11].

In this study, dietary intake data were obtained from 24-h recall interviews conducted in the Mobile Examination Center (MEC) as part of the NHANES project. We excluded supplements and medications, and used the average of two reliable 24-h recalls to estimate usual nutrient intake. Due to the limited variety of dietary components available in the NHANES database, we used 26 available components from the original 45 parameters for calculation. For components present in the original DII but missing from NHANES (i.e., the 19 unused parameters), we followed the methodological recommendations of the developers by treating them as having a "null" effect, meaning their intake was assumed to be equivalent to the standard global average intake. This approach does not compromise the validity of the final DII score [11].

The DII calculation was standardized based on global intake reference values. First, the Z-score for each dietary component was calculated for each participant as

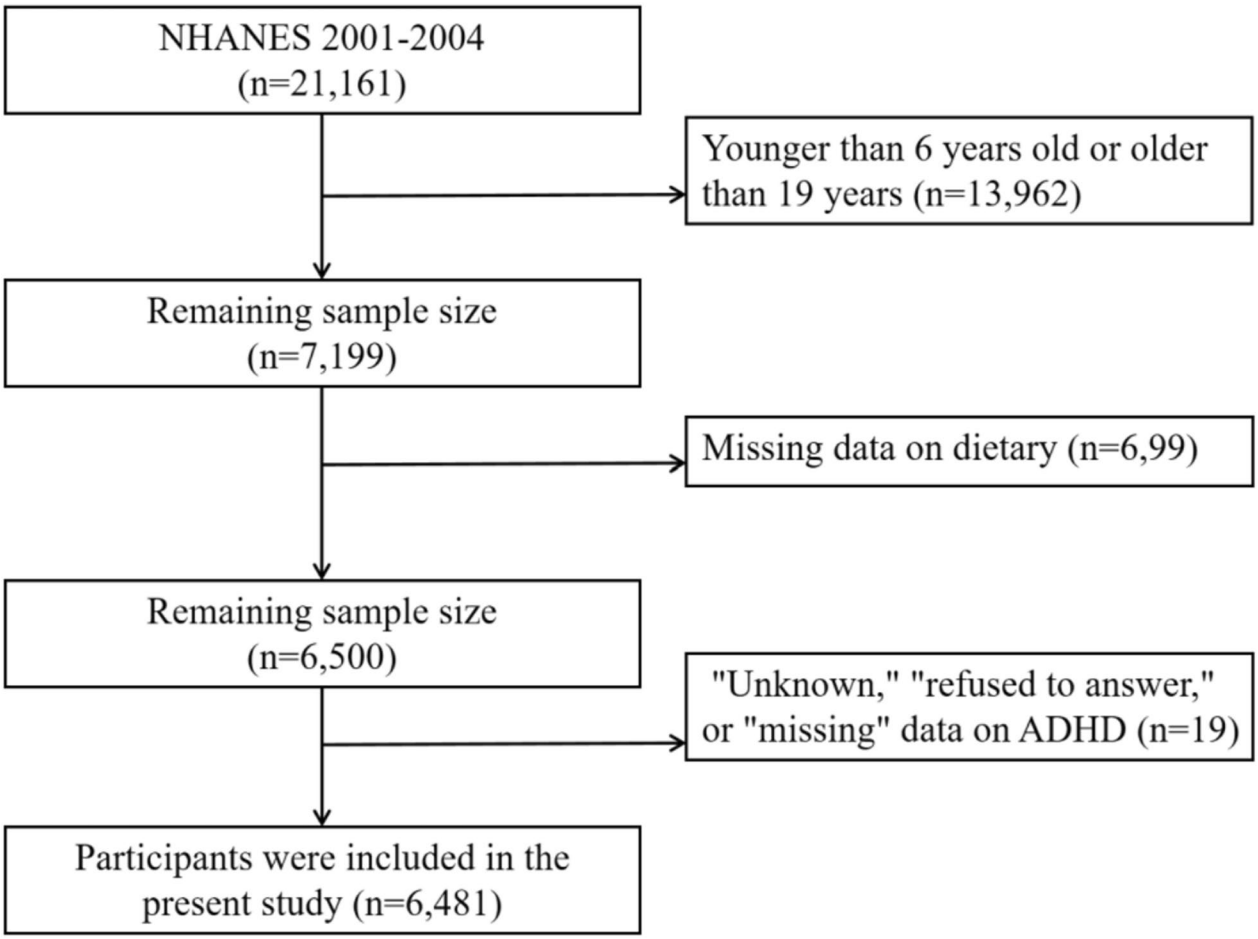


Fig. 1 Flow chart of sample selection from the NHANES 2001–2004

the difference between their daily intake and the global daily intake mean, divided by the global standard deviation (Z-score = [individual daily intake—global mean intake]/global SD). This Z-score was then converted to a percentile and further transformed to a score ranging from −1 to + 1 (Z-score' = percentile × 2−1). Finally, this score was multiplied by the respective component's inflammatory effect score (derived from the literature) to obtain the component-specific DII score. The overall DII score for an individual was the sum of all component scores. A positive total DII score indicates a pro-inflammatory dietary potential, while a negative score indicates an anti-inflammatory potential; the theoretical range of DII is from −8.87 (maximum anti-inflammatory potential) to + 7.89 (maximum pro-inflammatory potential). The DII was not energy-adjusted per 1000 kcal in this study, in accordance with the original DII calculation method and to facilitate comparison with other studies [11] (Table S1-2).

Measurement of indicators of Frailty Index (FI)

Frailty is increasingly recognized as a meaningful concept in pediatric populations, transcending mere physical vulnerability to encompass psychosocial and developmental dimensions. This broader understanding now informs clinical approaches for children with chronic conditions, disabilities, or complex healthcare requirements [21–23]. 49 health questions covering multiple systems make up the Frailty Index. These include one question about cognition, 15 questions about daily living activities, seven questions about mental health and depressive symptoms based on the patient's health questionnaire-9 questionnaire [24], 13 questions about chronic diseases, 5 questions about self-rated health, two anthropometric indicators, and six laboratory measurements (Table S2). On a scale of 0–1, each of these health issues has a score. According to Blodgett et al. [25], the information gathered from individuals who answered 30 or more items in a thorough health evaluation is considered enough to

precisely quantify negative effects. The ratio of the total number of health issues to the sum of the scores for those problems is the frailty score. Participants with greater frailty scores had a worse prognosis. Drawing on prior research, we established the cut-off point for the Frailty Index at 0.25. The FI was constructed following the established deficit-accumulation approach to serve as a composite indicator of overall health burden, consistent with its growing application in pediatric health research. Individuals with an FI score below 0.1 were classified as non-frailty, those with an FI score between 0.1 and 0.25 were categorized as pre-frailty, and those with an FI score of 0.25 or higher were classified as frailty [26].

Assessment of other variables

To gather information on demographics, lifestyle, self-reported health status, anthropometric measures, and biochemical markers, the CDC used a computer-assisted personal interview (CAPI) system. Demographic factors in this research were health insurance status, age, sex, and race/ethnicity. The socioeconomic situation of participants was represented by the poverty income ratio (PIR), which was the ratio of family income to the poverty criteria suitable for household size. Categories for the PIR were < 1.00, 1.00–1.84, 1.85–3.00, and > 3.00 [27]. Aggregate Index of Systemic Inflammation (AISI) [28] = (Platelet count × Neutrophil count × Monocyte count)/Lymphocyte count. A self-reported history of hypertension, self-reported use of antihypertensive drugs, or an average systolic or diastolic blood pressure that is higher than the 90th percentile for age, sex, and height were all considered indicators of elevated blood pressure (EBP). EBP was taken into consideration for children 13 years of age and older if their systolic or diastolic blood pressures were more than 120 mmHg or 80 mmHg, respectively [29]. Platelet count, neutrophil count, and lymphocyte count are combined to create the Systemic immunological-Inflammation Index (SII), an indication obtained from standard blood testing that evaluates the body's inflammatory and immunological condition. The following is the calculating formula: SII is equal to (neutrophil count × platelet count)/lymphocyte count [30].

Mendelian randomization

Data sources

MR analyses utilized publicly accessible genetic data. The GWAS summary statistics for food liking were derived from a cohort of 161,625 UK Biobank participants. Participants rated their preference for individual food items on a 9-point Likert scale ranging from "Extremely dislike" to "Extremely like." These preferences were further grouped into three categories: "Highly-palatable," "Acquired," and "Low-caloric" [31]. The data were collected via questionnaire, with full details available at: https://biobank.ndph.ox.ac.uk/showcase/showcase/docs/foodpref.pdf.

Basic characteristics of the 187 dietary factors included as exposures or outcomes in the Mendelian randomization analysis are provided in Table S10. ADHD GWAS data were obtained from the Psychiatric Genomics Consortium (https://pgc.unc.edu/), wherein cases comprised individuals diagnosed by psychiatrists using ICD-10 criteria (F90.0, F90.1, F98.8) or those receiving pharmacotherapy for ADHD symptoms [32].

Instrumental variable (IV) selection

Independent single-nucleotide polymorphisms (SNPs) serving as IVs for the food preferences were selected. Genome-wide significant SNPs ($p < 5 \times 10^{-8}$) were clumped for linkage disequilibrium (LD) using a reference panel (r^2 threshold < 0.001 within 1 Mb genomic windows). Two-sample MR analyses were subsequently conducted using these IVs and the ADHD GWAS summary statistics. For each of food preferences, SNPs serving as genetic instruments were rigorously screened. To ensure the robustness of genetic instruments, we excluded palindromic SNPs (i.e., those with A/T or G/C alleles) from the analysis to avoid potential strand ambiguity and misalignment between exposure and outcome datasets. Additionally, we aligned all SNPs to the forward strand based on the reference genome prior to harmonization.

Statistical analysis

Cross-sectional study

Sample weights were used in all analyses because of the NHANES' intricate survey methodology. We utilized combined data from NHANES 2001–2004, applying the 4-year interview weight (WTINT4YR) for overall analyses. For analyses involving dietary data (e.g., DII calculation), the average of two-day dietary recall weights ([WTDRD1 + WTDRD2]/2) was used to improve accuracy in estimating dietary exposures and outcomes. In R (v4.4.0), the survey package (v4.4–0) was employed to define the survey design, specifying: strata = SDMVSTRA, primary sampling units (PSUs) = SDMVPSU, and weights = WTINT4YR (overall analysis) or the average dietary weight (diet-related analysis). All regression models, including weighted logistic regression and restricted cubic spline (RCS) models, were fitted using the svyglm() function to ensure parameter estimation and hypothesis testing accounted for the complex survey design. In RCS analyses, svyglm() combined with rms::rcs() was used, with knot locations determined based on the weighted data distribution to ensure the spline fitting also

incorporated the sampling design. Continuous variables were described using the median and interquartile range (IQR), while categorical variables were represented using the unweighted frequency and weighted percentage. DII scores were divided into interquartiles for analytical reasons (T1: < 0.92, T2: 0.92–2.16, T3: 2.16–3.11, T4: > 3.11). With higher DII scores suggesting a larger inflammatory potential of the diet, the highest quartile (T4) reflects the most pro-inflammatory diet. On the other hand, the lowest quartile (T1), which has lower DII values, indicates a stronger anti-inflammatory diet. The Kruskal–Wallis H test or weighted chi-square was employed to ascertain differences between groups. Weighted logistic regression models were used to examine the relationship between DII scores and ADHD. For each DII score, odds ratios (ORs) and 95% CIs were calculated, using the lowest quartile as the reference. Using a restricted cubic spline (RCS), the likely non-linear relationship between ADHD and DII was shown. A total of three models were employed throughout the analysis. Model 2 adjusts for age, sex, race/ethnicity, household reference person's education level. Model 3 continues to adjust health insurance and PIR. Additionally, we conducted subgroup analyses and tested for interaction effects among the subgroup results (These analyses are interpreted as exploratory and hypothesis-generating, since they report subgroup-specific p-values without multiple-testing correction).

To explore the potential non-linear relationship between DII and ADHD, we fitted restricted cubic splines (RCS). The knots were placed at the following percentiles of the DII distribution: the 5th (−1.30), 35th (1.46), 65th (2.72), and 95th (3.92) percentiles, resulting in 3 degrees of freedom. The model formula was: logit (ADHD) = rcs(DII, 3) + covariates. The median DII value (2.16) was used as the reference for the unadjusted model. For the multivariable-adjusted model (adjusted for age, sex, race/ethnicity, household reference person's education level, PIR, and health insurance status), the median DII value (2.15) was used as the reference. All RCS analyses incorporated the NHANES complex survey weights to ensure the representativeness of the estimates for the US child and adolescent population.

To further quantify the inflection point of the non-linear association, we performed a piecewise linear regression analysis to identify a potential threshold effect. Using the segmented package in R, the breakpoint was identified through an iterative process that maximizes the model's likelihood function. The optimal breakpoint was identified at approximately DII 2.9. We then used a likelihood ratio test (LRT) to compare the goodness-of-fit between the two-piecewise linear model (with the breakpoint) and the simple linear model. The LRT chi-square statistic indicated that the two-piecewise model provided a significantly better fit to the data than the linear model ($P = 0.001$), confirming the presence of a threshold effect.

To explore the potential mediating role of the FI in the association between the Dietary Inflammatory Index (DII) and ADHD, we conducted a mediation analysis. The analysis was based on the counterfactual framework and proceeded in three steps: (1) fitting a linear regression model (svyglm with family = "gaussian") of DII on the mediator (FI); (2) fitting a logistic regression model (svyglm with family = "quasibinomial") of DII on the outcome (ADHD); and (3) fitting a logistic regression model of both DII and FI on ADHD. Formal mediation analysis was performed using the mediation package in R. To quantify the uncertainty of the mediation effect, we computed 95% confidence intervals (CIs) using the Quasi-Bayesian method based on 50 simulations. We first tested for the potential exposure-mediator interaction (DII × FI) by including an interaction term in the outcome model; as it was not statistically significant ($P > 0.05$), it was omitted from the final model. The proportion mediated (PM) and its 95% CI were estimated accordingly.

For variables with missing values, the zstats platform (www.zstats.net) performs multiple imputation with 5 imputations ($m = 5$), using predictive mean matching (PMM) for continuous variables and logistic regression imputation for categorical variables. All variables included in the analytical model are automatically incorporated into the imputation model. The imputation process convergence is ensured by the platform's internal algorithms, and the final results are pooled following Rubin's rules. All analyses were conducted bilaterally, with the statistical significance level for alpha established at 0.05. Data processing and analysis were conducted using R version 4.4.0 (2024–04–24) and Zstats 1.0 (www.zstats.net).

Mendelian randomization

MR leverages genetic variants as instrumental variables to infer causal relationships between exposures (food preferences) and outcomes (ADHD), contingent upon satisfying three core assumptions. We employed inverse-variance weighted (MR-IVW) analysis with a random-effects model as the primary analytical method to derive causal estimates for each exposure, which combines Wald ratio estimates from individual SNPs through meta-analysis. Comprehensive sensitivity analyses were conducted to validate MR assumptions, including MR-Egger regression (assessing directional pleiotropy), weighted median estimation (providing robustness against invalid instruments), and heterogeneity testing via Cochran's Q statistic. Crucially, the UK Biobank sample was explicitly excluded from this ADHD GWAS meta-analysis [31].

Therefore, our two-sample MR design is based on completely independent samples for exposures (UK Biobank) and outcome (PGC), effectively eliminating bias from sample overlap.

Results

Cross-sectional study

Features of the research population

A total of 6481 participants were included in this study, of whom 488 had ADHD, accounting for 7.53%, and 5993 were non-ADHD participants, accounting for 92.47%. Significant differences were observed between the two groups in terms of age, sex, race/ethnicity, SII, segmented neutrophil count, white blood cell count, AISI, Frailty Index, EBP, health insurance status, and DII levels ($P < 0.05$). No significant variations were seen in the education level of the household reference person and the PIR ($P > 0.05$). It is apparent that, in comparison to non-ADHD participants, those with ADHD were more often male, Non-Hispanic White, and had health insurance. Moreover, the higher Frailty Index values among ADHD participants may indicate a greater accumulation of health deficits or a higher risk of health issues, such as metabolic syndrome, functional impairments, developmental problems, and psychological and behavioral health concerns (Table 1). It is noteworthy that the significant negative correlations between ADHD and several indices closely related to inflammation and immune system activation—DII, SII, segmented neutrophil count, white blood cell count, and AISI, which further highlight the potential link between ADHD and systemic inflammation or immune dysregulation.

Association between DII and ADHD

We conducted a multivariate logistic regression analysis to examine the association between DII and ADHD. In the unadjusted model, the DII exhibited a significant inverse association with ADHD, such that each one-unit increment in DII corresponded to an 11% reduction in the odds of ADHD (OR = 0.89; 95% CI 0.82–0.96; $P = 0.008$). Subsequently, in Model 2 and Model 3, after adjusting for age, sex, race/ethnicity, health insurance status, PIR and household reference person's education level, the relationship between DII and ADHD remained significant and negatively associated (OR = 0.91; 95%CI 0.82–0.99; $P < 0.05$) (Table 2). When DII was stratified into quartiles (T1–T4) for analysis, we constructed three weighted logistic regression models to examine the association between DII and ADHD. The Model 1 (T3) and Model 2 (T3) demonstrated statistically significant associations between DII and ADHD ($P < 0.05$) (Table S3).

Figure 2 RCS demonstrates a U-shaped association between the DII and ADHD, providing empirical evidence for this nonlinear relationship (P for nonlinearity < 0.01). We consequently employed a piecewise logistic regression model to quantify the DII-ADHD association. Threshold effect analysis using a two-segment logistic regression model (Table 3) identified an optimal DII inflection point at approximately 2.9 (log-likelihood ratio test $P = 0.001$, confirming superior model fit compared to linear regression). Below this threshold, each unit increase in DII was associated with a 13% reduction in ADHD risk (OR = 0.87, 95% CI 0.81–0.94). Conversely, above the inflection point, each DII unit increment corresponded to a significant 161% elevation in ADHD prevalence (OR = 1.61, 95% CI 1.03–2.54), revealing differentiated dose–response patterns across DII ranges.

Subgroup analysis

Subgroup analyses were performed to explore potential effect modification. Importantly, all interaction terms were non-significant (interaction P-value > 0.05; Fig. 3), indicating no statistically robust evidence that the association between DII and ADHD differed across the subgroups examined. However, within specific strata, exploratory observations showed nominally significant negative correlations ($P < 0.05$) in participants with/without health insurance, among Other Hispanic individuals, and in those with low neutrophil counts, low SII/AISI scores, or pre-frailty/frailty status, both before and after adjusting for age, household education, and PIR (Table S4). Given the lack of significant interaction, these stratum-specific observations should be interpreted with caution and considered hypothesis-generating for future research, rather than as evidence of true subgroup differences.

Sensitivity analysis

Considering the potential confounding effects of medications on both Dietary Inflammatory Index (DII) values and inflammatory pathways, this analysis specifically evaluated the DII-ADHD association among medication-naïve participants. We excluded individuals reporting use within the preceding 30 days of analgesic/anti-inflammatory drugs (ibuprofen, acetaminophen, aspirin), glucocorticoids, lipid-lowering statins, psychotropic medications, and other relevant agents. The final analytical cohort comprised 5069 participants [27]. Consistent with the primary analysis, multivariable regression models (Model 1–3) were employed to assess the DII-ADHD relationship. Results confirmed that among children and adolescents aged 12–19 years, significant associations persisted for both continuous DII (without age

Table 1. Baseline characteristics of participants according to ADHD

Variable	Total ($n = 6481$)[1]	With ADHD (n=488)[1]	No ADHD (n=5993)[1]	Statistic	P[2]
Age (years), *n* (%)	12.44 (0.11)	12.90 (0.23)	12.39 (0.12)	t=−2.17	**0.038**
Sex, *n*(%)				χ^2=131.97	**<0.001**
Male	3227 (51.28)	359 (73.78)	2868 (49.02)		
Female	3254 (48.72)	129 (26.22)	3125 (50.98)		
Race/ethnicity, n (%)				χ^2=34.54	**<0.001**
Mexican American	1953 (11.67)	75 (5.31)	1878 (12.31)		
Other Hispanic	241 (6.13)	22 (6.69)	219 (6.07)		
Non-Hispanic White	1858 (61.65)	210 (70.60)	1648 (60.75)		
Non-Hispanic Black	2171 (14.73)	158 (12.41)	2013 (14.96)		
Other Race – Including Multi-Racial	258 (5.82)	23 (4.98)	235 (5.90)		
SII, Mean (SE)	529.45 (8.43)	491.56 (13.36)	533.27 (8.41)	t=3.87	**<0.001**
Segmented Neutrophils ($\times 10^3$ cells/μl), Mean (SE)	3.87 (0.04)	3.67 (0.07)	3.89 (0.05)	t=3.15	**0.004**
White Blood Cells ($\times 10^3$ cells/μl), Mean (SE)	7.10 (0.05)	6.85 (0.11)	7.13 (0.05)	t=2.90	**0.007**
AISI, Mean (SE)	319.92 (6.94)	293.41 (11.08)	322.59 (6.91)	t=3.35	**0.002**
Frailty Index, Mean (SE)	0.17 (0.00)	0.19 (0.01)	0.16 (0.00)	t=−3.32	**0.002**
EBP, *n*(%)				χ^2=26.07	**0.031**
No	3989 (84.30)	336 (77.50)	3653 (85.09)		
Yes	739 (15.70)	72 (22.50)	667 (14.91)		
DII, Mean (SE)	1.85 (0.05)	1.56 (0.11)	1.88 (0.05)	t=2.74	**0.010**
DII score categories[3], *n* (%)				χ^2=21.99	**0.044**
T1	1689 (24.99)	157 (31.89)	1532 (24.30)		
T2	1620 (24.98)	122 (24.80)	1498 (25.00)		
T3	1637 (25.02)	90 (18.99)	1547 (25.62)		
T4	1535 (25.01)	119 (24.32)	1416 (25.08)		
Health insurance status, *n* (%)				χ^2=18.91	**<0.001**
Yes	5173 (85.35)	435 (91.30)	4738 (84.75)		
No	1215 (14.41)	43 (8.68)	1172 (14.98)		
Household reference person's education level, *n*(%)				χ^2=24.86	0.153
Less Than 9th Grade	817 (7.83)	27 (4.85)	790 (8.13)		
9-11th Grade (Includes 12th grade with no diploma	1348 (14.50)	103 (18.10)	1245 (14.15)		
High School Grad/GED or equivalent	1562 (26.71)	124 (29.21)	1438 (26.47)		
Some College or AA degree	1603 (28.69)	139 (30.99)	1464 (28.47)		
College Graduate or above;	892 (21.80)	72 (16.76)	820 (22.30)		
PIR, n(%)				χ^2=3.75	0.681
<1.00	1928 (22.34)	122 (23.70)	1806 (22.20)		
1.00–1.84	1487 (20.89)	109 (21.54)	1378 (20.82)		
1.85–3.00	1097 (19.45)	85 (16.48)	1012 (19.75)		
>3.00	1631 (37.32)	143 (38.27)	1488 (37.23)		

SE: Standard Error; t: t-test, χ^2: Chi-square test

[1] Mean (SE); [2] Wilcoxon rank-sum test for complicated survey samples; chi-squared test with Rao & Scott's second-order adjustment; [3]DII quartile ranges. T1: low DII scores, T2: middle DII scores, T3: higher DII scores, T4: highest DII scores

Mean ± SE for continuous variables: *P* value was calculated by weighted linear regression

Model. % for categorical variables: *P* value was calculated by weighted chi-square test

adjustment) and the highest DII quartile (T4) with ADHD risk following full covariate adjustment, demonstrating robustness of this relationship (Table S5). RCS analysis further established a significant nonlinear association between DII and ADHD (P for nonlinearity < 0.01; Fig. S1).

Table 2. The association between DII and ADHD in children and adolescents

Variables	Model 1		Model 2		Model 3	
	OR (95%CI)	*P*	OR (95%CI)	*P*	OR (95%CI)	*P*
DII	0.89 (0.82–0.96)	**0.008**	0.90 (0.82–0.98)	**0.026**	0.91 (0.82–0.99)	**0.050**

Model 1: no covariates were adjusted; Model 2 was adjusted by sex, race/ethnicity, household reference person's education level; Model 3 was adjusted by sex, race/ethnicity, household reference person's education level, age, health insurance status, PIR

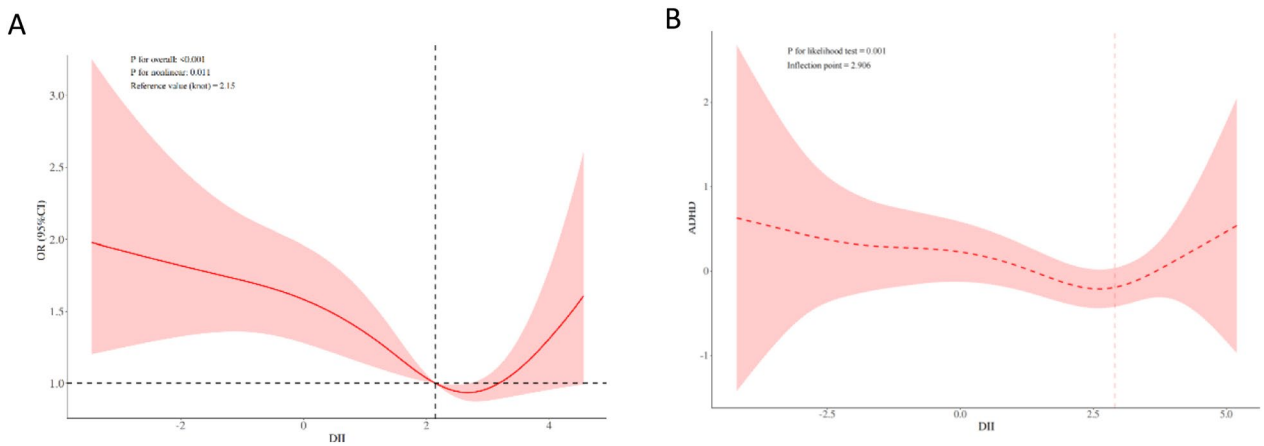


Fig. 2 **A** RCS analysis was used to examine the nonlinear relationship between the DII and ADHD. The model incorporated NHANES sampling weights and was adjusted for age, sex, race/ethnicity, household education level, health insurance status, and PIR. Confidence intervals were calculated based on the standard errors from the weighted logistic regression model. **B** Threshold effect analysis of the relationship between DII and ADHD. All analyses utilized NHANES sampling weights and were adjusted for the covariates listed above. The solid red line represents the odds ratio (OR), and the dashed lines represent the 95% confidence interval. ADHD, attention deficit hyperactivity disorder; DII, dietary inflammatory index

Table 3 Results of two-piecewise logistic-regression model

DII	ADHD, OR(95% CI)	*P**
Model 1 Fitting model by standard linear regression	0.95 (0.89–1.01)	0.078
Model 2 Fitting model by two-piecewise linear regression		
Inflection point	2.906	
< 2.906	0.87 (0.81–0.94)	**< 0.001**
≥ 2.906	1.61 (1.03–2.54)	**0.039**
P for likelihood test		**0.001**

* Model was adjusted for age, sex, race/ethnicity, household reference person's education level, health insurance status, and PIR

Mediation analysis

Considering that the association between DII and ADHD was strengthened by the inclusion of the Frailty Index as a covariate (as shown in Model 3 of Table 2), we proceeded to conduct a mediation analysis. Our observations indicated that the Frailty Index serves a partial mediating function between DII and ADHD in the analysis. The mediation model and the pathways are depicted in Fig. 4. The study results demonstrate a significant correlation between DII and the Frailty Index ($\beta = 0.002$, $P = 0.01$), as well as a significant positive correlation between the Frailty Index and ADHD ($\beta = 2.88$, $P < 0.001$). Further analysis confirmed that DII exerts a significant indirect influence on ADHD via the Frailty Index ($P < 0.001$), with an indirect effect size of 0.0006. This indicates that the Frailty Index mediates, to a partial extent, the relationship between DII and ADHD, constituting approximately 5.49% (−0.05, 95% CI −0.18 to −0.01) of the total effect. These findings suggest a potential mediating role of the Frailty Index in the observed association between DII and ADHD.

Mendelian randomization study

We employed five analytical approaches to evaluate causal relationships between food preferences and ADHD, utilizing inverse-variance weighted (IVW) as the primary MR method. Following false discovery rate (FDR) correction (FDR < 0.05), genetically predicted fruit liking (OR = 0.659, 95% CI 0.521–0.833, $P = 5.007 \times 10^{-4}$) and cheesecake liking (OR = 0.781, 95% CI 0.673–0.906, $P = 0.001$) were associated with reduced ADHD risk.

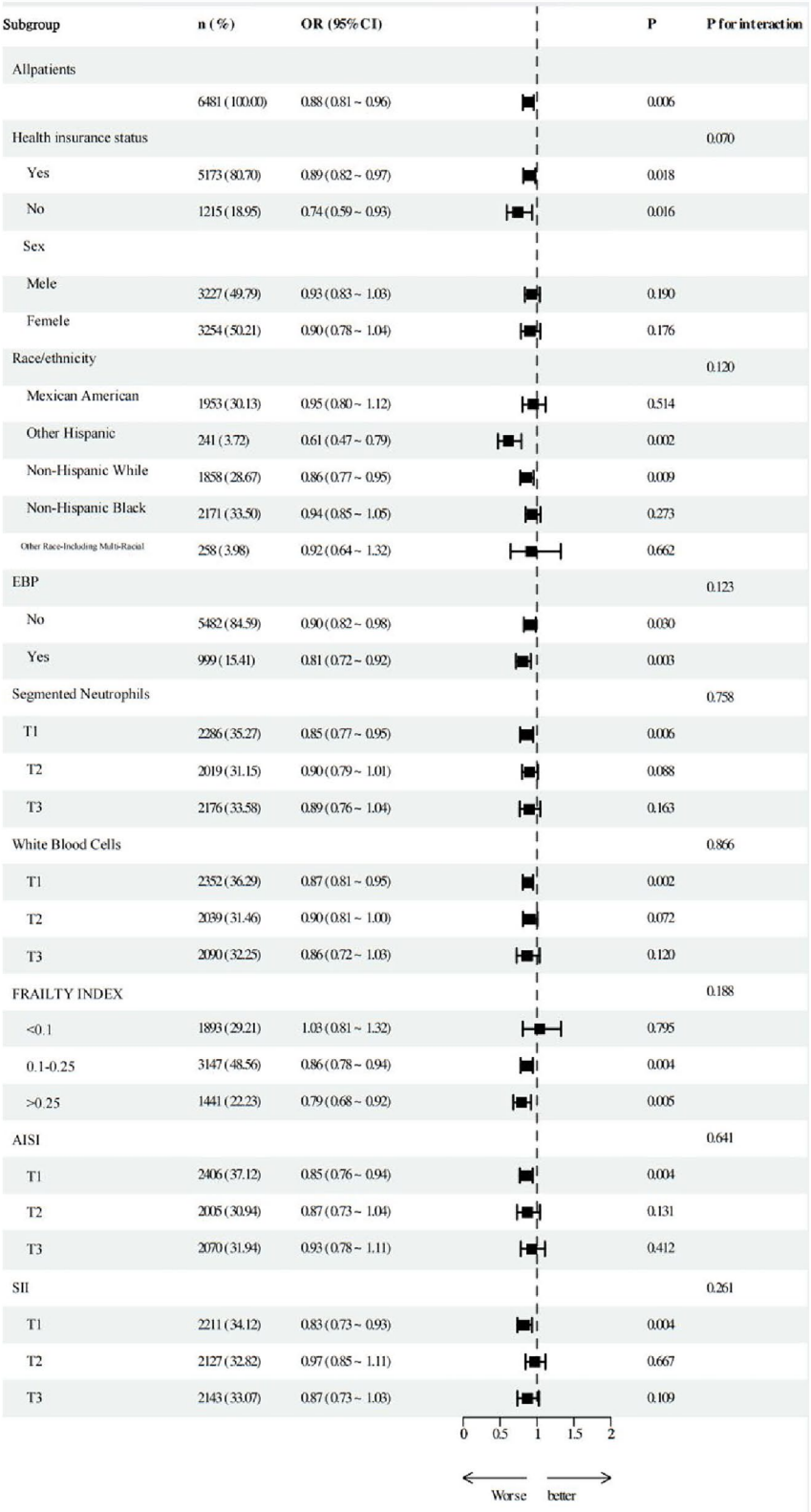


Fig. 3 Forest plot of the stratified analysis for the association between DII and ADHD. All analyses employed NHANES sampling weights and were adjusted for age, household education level, and PIR within each stratum. The AISI and SII were categorized into tertiles (T1–T3); the Frailty Index (FI) was classified as robust (FI ≤ 0.10), pre-frail (0.10 < FI < 0.25), and frail (FI ≥ 0.25). Confidence intervals were derived from the weighted logistic regression models. EBP, elevated blood pressure; AISI, Aggregate Index of Systemic Inflammation; SII, Systemic immunological-Inflammation Index

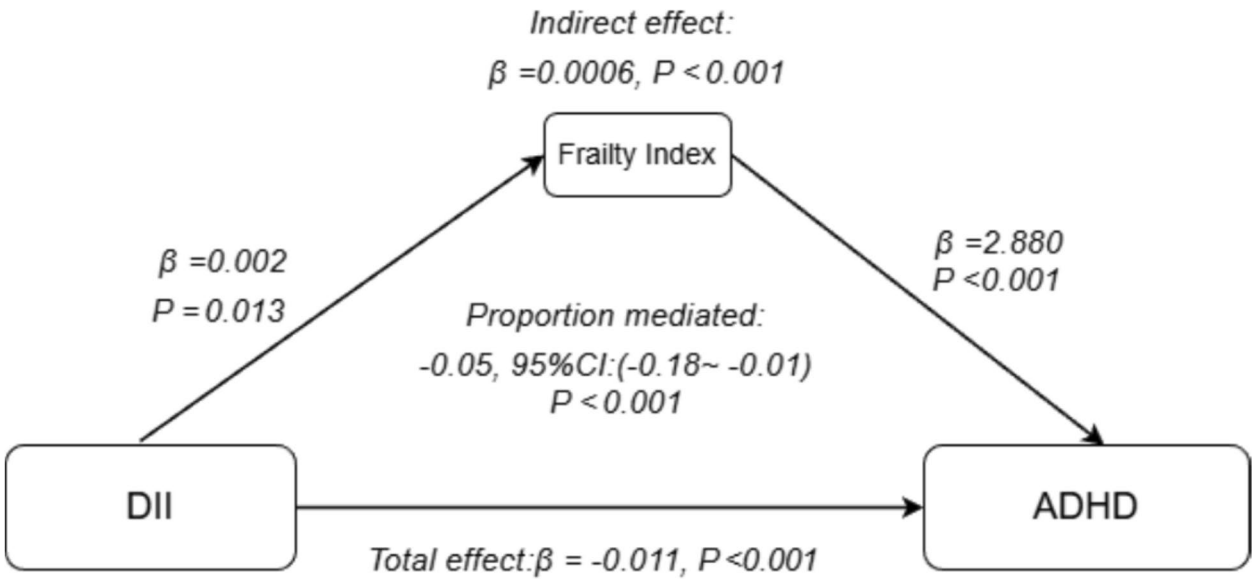


Fig. 4 Estimated fraction of the relationship between DII and ADHD mediated by the Frailty Index. ADHD, attention deficit hyperactivity disorder; DII, dietary inflammatory index

Table 4 Significant estimates for MR analysis

Exposure	Outcome	IVW-derived *p*-value	Beta	SE	Cochran's Q-derived *p*-value	MR-Egger intercept derived *p*-value	FDR
Fruit liking	ADHD	0.001	−0.417	0.120	0.694	0.817	0.029
Lamb liking	ADHD	0.001	0.342	0.100	0.216	0.199	0.029
Onion liking	ADHD	0.001	0.297	0.090	0.255	0.153	0.029
Cheesecake liking	ADHD	0.001	−0.248	0.076	0.517	0.183	0.029

ADHD, attention deficit hyperactivity disorder; IVW, inverse variance weighted

Conversely, preferences for lamb (OR = 1.408, 95% CI 1.157–1.714, $P = 6.467 \times 10^{-4}$) and onion (OR = 1.345, 95% CI 1.128–1.605, $P = 9.753 \times 10^{-4}$) were linked to increased ADHD risk in univariate MR analyses (Table 4). However, multivariable MR adjusting for correlated dietary traits showed that the association for lamb liking was attenuated and no longer significant (OR = 1.164, 95% CI 0.963–1.406, $P = 0.116$), indicating potential confounding by other dietary factors (Tables S9).

For all four univariable associations, MR-Egger intercept tests showed no evidence of directional pleiotropy ($P > 0.05$). MR-PRESSO detected no outliers ($P > 0.05$), and Steiger directionality tests supported the correct causal direction ($P < 0.05$). Heterogeneity was low (Cochran's Q $P > 0.05$). Instrument strength was sufficient across analyses (F-statistics 41–604), minimizing weak instrument bias. Together, these results support the robustness of the causal inferences (Table S6-7).

To investigate whether ADHD might influence the association between DII and ADHD observed in the cross-sectional study by altering dietary preferences, we further performed reverse two-sample MR analyses using ADHD as the exposure and 187 food preferences as outcomes. Our analysis revealed that genetic predisposition to ADHD significantly altered 37 food preferences (FDR < 0.05). Delving into the pattern uncovered an intriguing phenomenon: ADHD was associated with a generalized reduction in liking for a wide range of foods (accounting for 81.08% of the significant associations). Specifically, ADHD was associated with an increased liking for the following 8 food items or derived factors: BBQ/grilled meat, cherry, F-savour/caloric food liking, apple juice, F-seafood liking, sardines, aubergine, and salty pretzels. Conversely, ADHD was associated with a decreased liking for 29 food items or derived factors, including: F-highly palatable foods liking, sausages, dark chocolate, roast chicken, croissant, F-meat liking, fruit, avocado, tea difference liking, tea without sugar liking, grapefruit, chicken, F-sweet food liking, F-sweetened coffee drink liking, coffee max liking, F-strong vegetable liking, brussel sprout, F-coffee/alcohol liking, beef/steak, coffee with sugar liking, marzipan, fried chicken, F-deep

fried food liking, savoury biscuit, F-cooking vegetables liking, mackerel, F-small fish liking, and lamb liking. The robustness of all significant associations was supported by sensitivity analyses (MR-Egger intercept test $P > 0.05$; Steiger directionality test $P < 0.05$) (Table S9).

Discussion

This study reports for the first time, using NHANES data, a U-shaped association between DII and ADHD risk: When DII < 2.9, an increase in DII was associated with a reduced ADHD risk (OR = 0.87); whereas when DII ≥ 2.9, each 1-unit increase in DII significantly elevated ADHD risk by 61% (OR = 1.61). This nonlinear pattern suggests that both extremely anti-inflammatory and pro-inflammatory diets may be associated with ADHD risk, implying that the inflammatory potential of diet, at both extremes, might be a relevant factor to consider in future investigations into ADHD etiology. This U-shaped association remained significant after adjusting for all covariates (sex, race/ethnicity, education, PIR, age, health insurance) (P for nonlinearity = 0.01, log-likelihood ratio test $P = 0.001$), with an inflection point at approximately 2.9 (Fig. 2). Notably, analysis using DII quartile groups (T1–T4), which imposes linearity, likely diluted the nonlinear trend when adjusting for all covariates, resulting in no significant differences between groups; the Cochran-Armitage trend test further confirmed a nonlinear change in ADHD risk across DII quartiles (trend test $P = 0.139$) (Table S3). RCS analysis is capable of characterizing the true relationship between exposure and outcome without presuming a predefined functional form. Therefore, our key finding—that a diet with moderate inflammatory potential is associated with the lowest risk of ADHD, whereas both extreme anti-inflammatory and pro-inflammatory diets may increase the risk—is primarily informed by the results of the RCS analysis. This observation underscores the importance of employing nonlinear modeling approaches when investigating complex exposures such as diet.

Research on dietary patterns and ADHD has grown substantially in recent years [16, 33]. As a key environmental factor, diet may influence ADHD onset and development through multiple mechanisms [34, 35]. Inflammation is a feature in some, but not all, ADHD patients and may be influenced by ADHD medication and lifestyle factors including diet, smoking, and stress [7]. To elucidate potential causal links between dietary liking and ADHD, we performed MR analyses on 187 dietary likings. Results revealed that a genetic predisposition for fruit liking significantly lowered ADHD risk (OR = 0.659, 95%CI 0.521–0.833, $P = 5.007 \times 10^{-4}$). Studies report that fruits are rich in antioxidants (e.g., flavonoids, vitamin C) and fiber. Deficiencies in vitamins B, C, and E may increase vulnerability to oxidative stress abnormalities, as these antioxidants protect neuronal cell membranes from free radical damage [36]. Dietary fiber promotes probiotic proliferation, while pectin in fruits can bind neurotoxic heavy metals like lead and cadmium. Conversely, preferences for red meat (e.g., Lamb: OR = 1.408, 95%CI 1.157–1.714, $P = 6.467 \times 10^{-4}$) and condiments (e.g., Onion: OR = 1.345, 95%CI 1.128–1.605, $P = 9.753 \times 10^{-4}$) significantly increased ADHD risk. Diets high in red meat, processed foods, and refined carbohydrates are associated with elevated plasma inflammatory markers [37]. Red meat consumption (especially lamb, beef) is linked to risks of various psychiatric disorders (e.g., depression, schizophrenia) [38], potentially related to high levels of trans-unsaturated fatty acids in red meat. High saturated fat intake is also closely associated with reduced BDNF [39]. However, in multivariable MR analyses adjusting for correlated dietary traits, the association for lamb liking was attenuated and became non-significant, suggesting that its effect may be confounded by other dietary factors (such as a Western dietary pattern or processed food intake). This highlights the complexity of dietary exposures and the necessity of accounting for the co-varying structure of diet in causal inference. Studies show a significant association between a Western dietary pattern (high in takeaway foods, fast food, and red meat) and elevated CBCL scores in children, reflecting both internalizing and externalizing behaviors. In contrast, a healthy dietary pattern featuring nuts and low-energy-dense foods is associated with improved behavioral scores [40]. The association between genetically predicted cheesecake liking and lower ADHD risk was unexpected and contrasts with most observational data. This finding requires replication and should be interpreted with extreme caution; it may reflect unmeasured confounding or shared genetic influences rather than a protective effect of cheesecake per se. Recent MR studies suggest causal associations between seven inflammatory cytokines (CD40, IL-18, MMP10, bNGF, MCSF, FGF19, AXIN1) and ADHD. For instance, mast cells may induce neuronal damage via CD40, while Roseburia spp. increase ADHD risk indirectly by elevating MCSF (mediation effect 3.7%) [4, 41]. Diet can induce functional changes beyond the gastrointestinal tract (including the brain) by modulating interactions between microbiota and local immune cells, altering cytokine release into systemic circulation, or regulating immune cell distribution [42]. Dietary preferences can modulate gut microbiota, transmitting inflammatory signals to the brain via the vagus nerve. Imbalances from antibiotics, poor diet, or other factors can lead to dysbiosis, elevating levels of cytokines like TNF, IL-6, and IFN-β, causing inflammation that increases intestinal permeability

("leaky gut" syndrome) [43, 44]. These findings suggest dietary changes may impact ADHD risk, but this effect is likely modulated by multiple factors, including genetic background, lifestyle, and environmental factors. Future research will involve deeper mechanistic studies, such as using neuroimaging (e.g., PET scans) to detect CNS inflammatory marker expression or analyzing cerebrospinal fluid samples to directly measure CNS inflammatory markers. Additionally, animal models can simulate diet-induced inflammation effects on the CNS, with histological and molecular biology methods used to observe CNS inflammation changes, further validating the impact of diet and CNS inflammation on ADHD risk.

Mediation analysis indicated that the Frailty Index may partially mediate the association between the DII and ADHD, accounting for 5.49% of the total effect. Nevertheless, the cross-sectional design precludes causal inferences or delineation of temporal precedence; consequently, these findings should be regarded as a preliminary hypothesis to be tested in future longitudinal investigations. Moreover, although the Frailty Index was constructed with a standardized protocol, its interpretation in a predominantly healthy pediatric cohort warrants caution, and subsequent studies should validate this metric against long-term neurodevelopmental outcomes [23, 25]. Although the 5.49% effect size is statistically significant, suggesting FI is a potential pathway, other unmeasured mechanisms (e.g., neuroinflammation, gut-brain axis regulation) likely play important roles [4, 45]. To validate this pathway, future studies should employ prospective cohorts to track temporal changes in DII, FI, and ADHD symptoms; use animal models to investigate molecular mechanisms of DII regulating FI (e.g., effects of inflammatory cytokines IL-6/TNF-α on neurodevelopment); and integrate multi-omics technologies (epigenetic/metabolomic) to dissect the DII-FI-ADHD biological pathway.

Baseline analysis showed children with ADHD had significantly lower levels of inflammatory markers including white blood cells, neutrophils, AISI, and SII ($P < 0.05$). At first glance, lower levels seem counter to a simple pro-inflammatory hypothesis. However, existing literature on inflammatory markers in ADHD is inconsistent [46, 47], suggesting immune dysregulation may not manifest as uniform elevation across all markers or individuals. Our finding of an association between lower DII scores and ADHD diagnosis in baseline comparisons can be understood in this context of complexity. This phenomenon may partly stem from limitations of linear modeling. RCS analysis revealed a significant U-shaped association (P for nonlinearity < 0.01): the negative association at the low DII end (DII < 2.9) and the positive association at the high DII end (DII ≥ 2.9) together form the complete nonlinear pattern. This negative association might also reflect ADHD-diagnosed children adhering to higher quality diets. Furthermore, using the average of two 24-h recalls for DII calculation introduces potential bias due to representativeness issues of transient diet. Residual confounding (e.g., family dietary environment, exercise habits) may persist despite adjusting for multiple covariates.

We also evaluated the possibility of reverse causation—where an ADHD diagnosis might influence dietary behavior—using reverse MR analysis. The results indicated that genetic predisposition to ADHD was significantly associated with alterations in 37 food preferences, manifesting as a generalized reduction in preference for both pro-inflammatory foods (e.g., lamb, sausages) and anti-inflammatory foods (e.g., fruit, avocado). This pattern suggests that genetic liability to ADHD may influence appetite or food preferences through biological pathways (e.g., reward system dysfunction). The causal effect of ADHD genetic liability on food preferences, as suggested by reverse MR, provides a plausible biological mechanism that could partly explain the observed cross-sectional association between ADHD diagnosis and lower DII scores. However, this does not preclude other interpretations. Interestingly, this finding parallels a birth cohort study [48] which found higher maternal DII was associated with significantly reduced risks of inattention (OR = 0.86; 95% CI = 0.77–0.96), hyperactivity (OR = 0.82; 95% CI = 0.72–0.92), and overall ADHD risk (OR = 0.82; 95% CI = 0.72–0.93) in boys, suggesting a sex-specific negative association. However, other studies found prenatal pro-inflammatory diets (e.g., Western pattern) significantly increased offspring ADHD risk: three large mother–child cohorts confirmed associations with ADHD/autism, and metabolomic analysis identified 15 mediating metabolites during pregnancy that improved ADHD prediction [49]. Researchers also found that higher prenatal unhealthy diet scores were associated with methylation of the insulin-like growth factor 2 (IGF2) gene at birth in children diagnosed with ADHD or conduct disorder. They concluded that higher IGF2 methylation predicted ADHD symptoms and suggested preventing unhealthy diets during pregnancy could reduce childhood ADHD prevalence [50]. While reports of U-shaped associations between extremely low maternal DII and offspring ADHD are scarce, substantial evidence indicates that severe deficiencies of key nutrients during pregnancy (e.g., iron, iodine, folate, vitamin D, Omega-3 DHA) significantly increase risks of neurodevelopmental delays and cognitive/behavioral problems (including ADHD-like symptoms) [51]. Extreme restrictive diets causing nutritional inadequacy can alter maternal immunity, milk, and serum metabolomic profiles, and induce widespread changes in fetal brain transcriptomics

and metabolomics, including subsets of microbiome-dependent metabolites. Such diets are recognized as potentially increasing adverse pregnancy outcomes, and their long-term effects on offspring neurodevelopment warrant attention [51]. This highlights the need for balanced and adequate prenatal nutrition: avoiding highly pro-inflammatory diets while ensuring all essential nutrients are obtained for fetal neurodevelopment. Future research should explore the dose–response relationship between prenatal DII and offspring neurodevelopment more meticulously, particularly investigating whether risks increase at the low DII end due to nutritional insufficiency.

Furthermore, baseline analysis showed elevated blood pressure was significantly positively associated with ADHD, potentially linked to ADHD medication use and increased cardiovascular disease (CVD) risk [52]. Although most studies have relatively short follow-up periods, limiting assessment of long-term associations, some suggest no direct link, or a time-varying link, between ADHD and elevated blood pressure in children and adolescents [53]. Notably, the U-shaped relationship (Fig. 2) might underlie the lack of significance in some subgroup analyses. ADHD risk increases at both extremely low DII (strong anti-inflammatory) and high DII (strong pro-inflammatory) values. In subgroups like race/insurance, differences in the range of DII distribution may mask the nonlinear effect of the continuous variable. For instance, the median DII (2.218) in the "Other Hispanic" group is closer to the U-curve trough (inflection point ≈ 2.9), making the protective association easier to detect. Regarding healthcare access, children with ADHD were more likely to have health insurance. However, the negative association between DII and ADHD remained significant after adjusting for insurance status, and subgroup analyses showed consistent effects in both insured and uninsured groups (interaction $P > 0.05$), indicating that this association is unlikely to be entirely driven by differences in healthcare accessibility. Nevertheless, residual confounding (e.g., by family health awareness) cannot be fully ruled out.

This study has several notable strengths. First, all participants were directly sourced from the NHANES database, greatly enhancing sample representativeness and the generalizability of our conclusions. Second, we adhered to NHANES analysis standards, performing all analyses using appropriate sampling weights. Employing a stratified multistage probability sampling design, our results are generalizable to the US national population of children and adolescents. Third, while DII associations with other disorders exist, this is the first study to report a U-shaped association between DII and adolescent ADHD (inflection point ≈ 2.9). However, limitations must be acknowledged. First, the cross-sectional design precludes establishing a causal relationship between DII and ADHD risk. Second, all dietary information was self-reported (based on the average of two 24-h recalls), potentially introducing reporting bias, and the lack of long-term dietary data limits our ability to capture habitual dietary patterns. Differences in medical accessibility may introduce selection bias. Third, despite controlling for confounders via multivariable adjustment and sensitivity analyses, residual confounding (e.g., family dietary environment, exercise habits) may persist. Fourth, the inflammatory markers measured (e.g., WBC, neutrophils, AISI) reflect peripheral inflammation and cannot be directly equated to central nervous system (CNS) inflammation; future studies require validation via CSF analysis or neuroimaging (e.g., PET). Fifth, as NHANES primarily reflects the US population, subgroups with unique dietary patterns or health issues may be underrepresented, potentially limiting the generalizability of our findings to more diverse or international populations. Sixth, the use of a non-energy-adjusted DII, while consistent with the original index design and facilitating comparison with much of the literature, may introduce some confounding when comparing individuals with substantially different total energy intakes. Future studies could consider using the energy-adjusted DII (E-DII) to further verify the robustness of these findings. Furthermore, this analysis reveals a causal association between genetically predisposed "liking tendencies" and ADHD risk. This relationship may be partially mediated by actual dietary behaviors but could also reflect shared neurobiological pathways, such as those involving the reward system. Therefore, the results should be interpreted with caution. Their primary value lies in providing genetic-level support for a causal link between diet and ADHD, generating hypotheses for further investigation.

To address these limitations and rigorously test the hypothesis that 'moderate dietary interventions may help prevent ADHD', we propose a multi-level research agenda. First, there is a need to establish large-scale prospective birth cohorts with repeated dietary assessments from pregnancy through adolescence to confirm the U-shaped relationship and delineate critical exposure windows. Second, the 'gold standard' for establishing causality would be randomized controlled trials evaluating the efficacy of anti-inflammatory dietary patterns (e.g., a modified Mediterranean diet) in preventing ADHD onset or mitigating its symptom severity, particularly in high-risk populations. Finally, future research should integrate multi-omics approaches—including detailed measurements of inflammatory biomarkers, gut microbiota, and neuroimaging—to elucidate the precise biological mechanisms underpinning the DII-ADHD association, such as

the gut-brain axis and neuroinflammation. Such mechanistic insights are crucial for developing targeted and effective nutritional strategies for ADHD prevention and management.

Furthermore, although our mediation analysis indicates that the FI partially mediates the association between the DII and ADHD, it is important to note that the FI is a multidimensional construct encompassing cognitive, physical, psychological, and metabolic domains, comprising a total of 49 health deficits. Our analysis did not evaluate which specific components of frailty—such as impaired cognitive function, depressive symptoms, or limitations in daily activities—exhibit the strongest association with ADHD risk. Future studies with larger sample sizes and longitudinal designs could develop mediation models based on specific deficit clusters or distinct dimensional pathways to identify the most critical mechanistic routes—for instance, neurocognitive deficits or immunometabolic dysregulation—through which diet-related inflammation may influence the development of ADHD. Such findings would help inform more targeted interventions for at-risk pediatric populations. Additionally, the generalisability of adult-derived genetic variants underlying food preferences to paediatric populations and to diverse dietary-cultural contexts—particularly the United States—remains equivocal, as large-scale genome-wide association studies of food preferences have yet to be conducted in children. Although the present study employed the best-available data, future investigations leveraging child-specific genetic repositories are imperative to corroborate the putative causal pathways.

Conclusion

In US children and adolescents, we observed a U-shaped association (inflection point ≈ 2.9) between the DII and ADHD risk: moderate DII is protective, while extremely low or high DII may increase risk. The Frailty Index may partially mediate (5.49%) the association between DII and ADHD. Fruit liking lowered ADHD risk, while red meat/onion liking increased risk. However, the cross-sectional design precludes causal inference. Future research should validate the preventive potential of dietary interventions through birth cohorts or randomized controlled trials (RCTs).

Contributions to the literature

First large-scale evidence: This is the first nationally representative study (using NHANES data) to investigate the relationship between DII and ADHD diagnosis in US children and adolescents; Reveals complex relationship: It uncovers a novel U-shaped association between DII and ADHD risk, indicating that both highly anti-inflammatory and highly pro-inflammatory diets are linked to increased ADHD probability, challenging simple linear assumptions; Identifies a new pathway: The study provides initial evidence that the Frailty Index (a measure of accumulated health deficits) partly explains (mediates) the link between dietary inflammation and ADHD, suggesting a potential biological mechanism; The findings from the MR analysis revealed that fruit liking is significantly linked to a reduced risk of ADHD. Conversely, a liking for red meat, such as lamb, as well as a fondness for flavorings like onions, is significantly associated with an increased risk of ADHD. ADHD was associated with a generalized reduction in liking for a wide range of foods (accounting for 81.08% of the significant associations). Informs prevention: These findings highlight the potential importance of moderate dietary interventions targeting inflammation for ADHD prevention strategies in children.

Abbreviations

ADHD	Attention deficit hyperactivity disorder
DII	Dietary inflammatory index
NHANES	National health and nutrition examination survey
RCS	Restricted cubic spline
NCHS	National center for health statistics
CDC	Centers for disease control and prevention
CAPI	Computer-assisted personal interview
FI	Frailty Index
EBP	Elevated blood pressure
AISI	Aggregate Index of Systemic Inflammation
SII	Systemic immune-inflammation index
PIR	Poverty-income ratio
OR	Odds ratio
CI	Confidence interval
MR	Mendelian randomization
GWAS	Genome-wide association study
SNPs	Single-nucleotide polymorphisms
IV	Instrumental Variable
IVW	Inverse variance weighted

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s40001-025-03711-z.

Supplementary Material 1

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Author contributions

Idea and planning: Jue Hu, Jingying Ma, and Fadan Tang; Data gathering and management: Yulu Pan, Ying Wang and Jiaye Xu; Analysis and explanation of data: Bohan Wang and Xiaobo Xuan; Drafting of the paper: All authors; Final endorsement of the paper: All authors.

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Data availability
The datasets for this study can be found in the NHANES database (https://wwwn.cdc.gov/nchs/nhanes/Default.aspx).

Declarations

Ethics approval and consent to participate
In accordance with the Declaration of Helsinki, the national center for health statistics (NCHS) Ethics Review Board of the centers for disease control and prevention (CDC) approved the human subjects' investigations. Given that this study involved secondary analysis of publicly available data without personal identifiers, it was exempt from further ethical review and approval. Written informed consent was obtained from the patients/participants involved in this research.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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